



Database Management Systems

Week 7, Day 1

MongoDB—Aggregate Functions

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Nested Group BY

```
> use agree
< switched to db agree
> db.persons.aggregate([{$group: {
  _id: {
    age: "$age", gender: "$gender"}
  }
}])
< {
  _id: {
    age: 22,
    gender: 'female'
  }
}
```

\$count

```
> db.persons.aggregate([
  {
    $count: "allDocumentsCount"
  }])
< {
  allDocumentsCount: 1000
}
agree>
```

\$project

- Includes, Excludes or adds new field(s)

```
{ $project: { <field1>: <1>, <field2>: <0>, <newField1>:  
<expression> ... } }
```

- Examples

```
{ $project: {name: 1, "company.title": 1} }
```

\$project Example

```
> db.persons.aggregate(  
  [  
    {  
      $project: {name: 1, _id: 0}  
    }  
  ]  
)  
< {  
  name: 'Aurelia Gonzales'  
}
```

\$project Example

```
db.persons.aggregate([
  {
    $project: { name: 1, newAge : "$age"}
  }
])
{
  _id: ObjectId('675fa2a3be57a37defbdcab3'),
  name: 'Aurelia Gonzales',
  newAge: 20
}
```

\$project Examples

```
> db.persons.aggregate([
  {
    $project: { name: 1, "company.location.country": 1 }
  }
])
{
  _id: ObjectId('675fa2a3be57a37defbdcab3'),
  name: 'Aurelia Gonzales',
  company: {
    location: {
      country: 'USA'
    }
  }
}
```



```
> db.persons.aggregate([
  {$limit : 10},
  {$match : {age : {$gt: 20}}},
  {$group : {_id : "$company.location.country"}}
])
< {
  _id: 'USA'
}
```

\$limit

```
persons.aggregate([
  limit : 100},
  match : {eyeColor: {$ne : "blue"}}},
  group : {_id : {eyeColor : "$eyeColor", favoriteFruit: "$favoriteFruit"}},
  sort: {"_id.eyeColor":1, "_id.favoriteFruit":-1}}

d: {
  eyeColor: 'brown',
  favoriteFruit: 'strawberry'}
```

\$limit Examples

Question(s)
